Python Quick Start Guide for Windows: Mech. Sensors Traner

STEP 1 Check Components and Details

Make sure your Mechatronic Sensors Trainer case includes the following components.

1. Mechatronic Sensors Trainer
2. Base
3. USB A – C Cable
4. USB C – C Cable
5. Force Sensing Resistor
6. Thermocouple
7. Load Cell and Scale Pieces
8. Radar Cone
9. Sensors Trainer Resources. Content and courseware provided in digital form at www.quanser.com/resources

Ensure you local computer has the following components

1. Windows 10/11 operating system
2. Python 3.11, 3.12 or 3.13 added to PATH
3. Quanser SDK 2026 or QUARC 2026 or later for either, installed

STEP 2 Download Content and Set Up Computer

A

Download the resources for the device from
https://github.com/quanser/Quanser_Academic_Resources
Follow the instructions on the 'Downloading Resources' and 'Setting Up Your Computer' sections.

B

You should now have Python 3.11, 3.12 or 3.13 downloaded and the 'Completing the Setup' section from 'Setting Up Your Computer' should have required a restart of your PC after installing all necessary libraries.

C

Download VS Code or a code editor of your choice. You should now have all the necessary files to run the device.

STEP 3 Set Up the Hardware

The steps below outline the instructions to setup the Mechatronic Sensors Trainer.

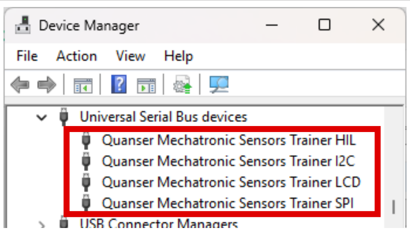
A

Connect one of the provided USB C connector to the device and then to an enabled USB 3.0 (or higher) port on your computer.

The power LED next to the connector should turn to white and the LCD screen should load the Quanser Logo and then show a black background with white crosses in a grid. LCD will say Low-power mode if USB A to C was used. That is expected, see the user manual for more information. .

If step 2 was done successfully and QUARC or Quanser SDK were properly installed, Windows[1] should automatically detect the presence of the Sensors Trainer.

B



Go to Windows Device Manager and verify that the device appears as multiple devices (HIL, I2C, LCD and SPI) under *Universal Serial Bus devices*.

STEP 4 Testing the Mechatronic Sensors Trainer

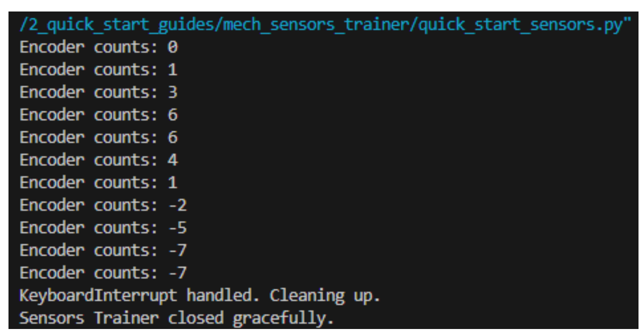
Follow the procedure below to test your Sensors Trainer.

A

Open **`quick_start_sensors.py`** on your code editor (VS Code) and click run.

B

1. This code is meant to print the encoder counts in the terminal every 0.5 seconds.
2. If it fails while initializing and loading any library, it will print instructions on how to set up the system.
3. If it starts printing, rotate the encoder knob next to the LCD screen and observe the counts change. Your device should be ready to use!
4. The code automatically stops after 200 seconds, or you can stop it using Ctrl+C on the terminal.



TROUBLESHOOTING

Review the following recommendations before contacting Quanser's technical support engineers.

1. Check the connections outlined in Step 3 of this guide and ensure that the cables are connected firmly.
2. If the device is turned on properly and the error happens when running the python code, follow the instructions printed on the terminal to understand what could have gone wrong.

LEARN MORE

To browse and download the latest Quanser resources visit https://github.com/quanser/Quanser_Academic_Resources

STILL NEED HELP For further assistance from a Quanser engineer, contact us at tech@quanser.com

[1] Windows is a registered trademark of Microsoft Corporation in the United States and other countries.